

SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:

Learn about the different types of salt on Earth. Then, learn about the origins of bones. In health, tackle the silent epidemics that are plaguing the world and finally, in R&D, take a look at how XR is changing science.



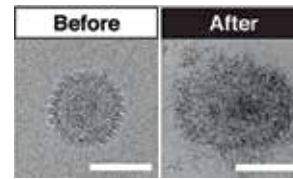
Paper trial

A fast test to diagnose the novel Coronavirus is the need of the hour, and a startup spunout from tech developed by MIT may have just the thing. The team has developed a low cost paper based test that they claim can provide results in under 30 minutes, and are now hoping to test patient samples. The approval for the test may get fast tracked because of a provision that allows faster authorisation in times of emergencies. <https://dgit.in/pprtst>

WHAT'S NEW

Scientists develop novel antiviral treatment based on sugar

A team of British and Swiss researchers have developed a new sugar based treatment that destroys viruses and can help during epidemics or pandemics. The new substance is a virucide, something that can totally annihilate viruses. There are known virucidal substances, such as bleach, but these are typically extremely harmful for use in the human body. For humans, treatments typically involve antiviral drugs, or vaccines. However, viruses mutate rapidly, and the coronavirus has a long history of evolving into virulent and deadly strains. Mutated viruses can evolve to be resistant to the drugs and vaccines.



The sugar based molecule breaks down the protective shell around the virus on contact, making

it hard for the virus to evolve resistances. A sugar based virucide can safely be used on humans as well. Apart from just destroying the virus, the molecule also attracts the virus to itself. This is a broad spectrum treatment, and the same molecule can tackle a number of viruses.

<https://dgit.in/killswitch>

Radiocarbon dating gets fine tuned

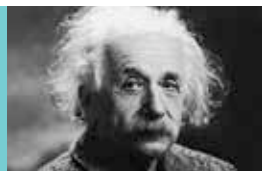
Since its discovery in the 1940s, radiocarbon dating has become the standard way of figuring out how old archaeological findings are. The method works for any material containing organic matter. To date a find, scientists measure the amount of carbon-14 decomposition, which is an unstable isotope of carbon. The isotope is created by cosmic radiation and deposited on all organic matter. However, the amount of cosmic radia-



tion received by the Earth is not constant. To calibrate radiocarbon dating, a series of tree rings are used. Tree rings are dated to the present day, and then overlapping reference slices go back all the way up to 50,000 years. There

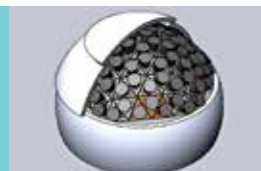
is a single calibration curve for the Northern Hemisphere, which was standardised in 1986. Now, researchers have found out that it is inaccurate to have a single reference for the northern hemisphere, as the trees in Europe or North America removed carbon from the atmosphere at different times of the year than the trees in Israel or Jordan. The slight difference can move historical events by decades.

<https://dgit.in/betterdating>



Einstein didn't work alone

According to historians working with the personal papers of Einstein, the eccentric looking genius did not work alone in isolation out of his attic, but collaborated extensively with his colleagues, much like the big science projects of today. <https://dgit.in/lonstein>



Fly's eye on the sky

Researchers from Caltech are building a pair of telescopes, made up of an array of imaging instruments placed on a geodesic dome, similar to a fly's eye. The purpose is to hunt for short lived laser pulses from civilisations trying to message Earth. <https://dgit.in/flyeye>



Softwire for the brain

Stanford researchers have created a new nanowire based brain computer interface that can be safely inserted into the brain for long periods of time, and allow scientists to measure neural activity in real time. The implant can help in brain research and prosthetics. <https://dgit.in/softwire>